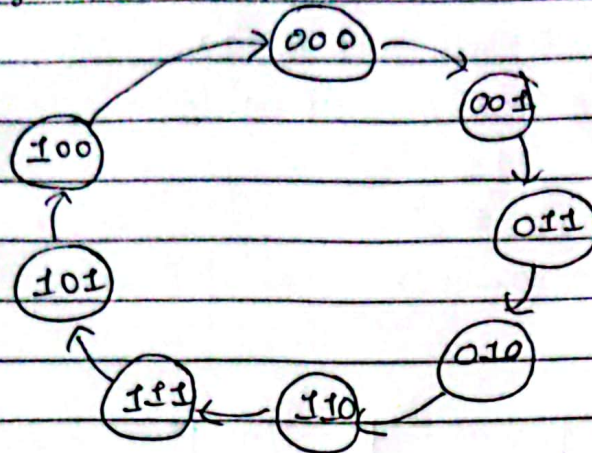


* Design 3 bit gray code counter (synchronous)

⇒ Solution:-

1) State diagram



Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

2) State Table.

Present state			Next state			Flip Flops		
						FF ₂	FF ₁	FF ₀
Q ₂	Q ₁	Q ₀	Q ₂ '	Q ₁ '	Q ₀ '	T ₂	T ₁	T ₀
0	0	0	0	0	1	0	0	1
0	0	1	0	1	1	0	1	0
0	1	1	0	1	0	0	0	1
0	1	0	1	1	0	1	0	0
1	1	0	1	1	1	0	0	1
1	1	1	1	0	1	0	1	0
1	0	1	1	0	0	0	0	1
1	0	0	0	0	0	1	0	0

3) K-map

T ₂					T ₁					T ₀				
Q ₁ Q ₀					Q ₁ Q ₀					Q ₁ Q ₀				
Q ₂	00	01	11	10	Q ₂	00	01	11	10	Q ₂	00	01	11	10
0				1	0		1			0	1		1	
1	1				1			1		1		1		1

$$T_2 = \bar{Q}_2 Q_1 \bar{Q}_0 + Q_2 \bar{Q}_1 \bar{Q}_0$$

$$= (Q_2 \oplus Q_1) \bar{Q}_0$$

$$T_1 = \bar{Q}_2 \bar{Q}_1 Q_0 + Q_2 Q_1 Q_0$$

$$= (Q_2 \oplus Q_1) Q_0$$

$$T_0 = \bar{Q}_2 \bar{Q}_1 \bar{Q}_0 + \bar{Q}_2 Q_1 Q_0 + Q_2 \bar{Q}_1 Q_0 + Q_2 Q_1 \bar{Q}_0$$

$$= Q_2 \oplus (Q_1 \oplus Q_0)$$

